12.22 A Distribution With No Mean

WORKSHEET

The question

STEP III 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If (U, V) is a random point whose density depends only on its distance from the origin, what can you say about the angle Θ that the line from the origin to the point makes with an axis?

STAGE 2 Using the symmetry

Ask yourself $\frac{U}{V} < k$ happens in two ways, according to the sign of V. How are they related by the rotation through $180°$, $(u, v) \mapsto (-u, -v)$?

STAGE 3 An angle

Ask yourself Draw the region $\{V > 0, U < kV\}$ in the plane with u horizontal and v vertical. It is a wedge with its vertex at the origin. What is its angle?

Try this next

1 **The torch.** A torch at distance 1 from a long straight wall points in a direction making angle Θ with the perpendicular, where Θ is uniform on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Show that the spot's position $\tan\Theta$ has the Cauchy distribution.

2 **The reciprocal.** Show that if C is Cauchy, so is $\frac{1}{C}$. (*Hint:* $\frac{U}{V}$ and $\frac{V}{U}$.)

3 **The median works.** Show that the median of the Cauchy distribution is 0 and its quartiles are ± 1.

4 **Radius and angle.** For X, Y independent standard normals, show by changing to polar coordinates that $R^2 = X^2 + Y^2$ is exponential with mean 2 and is independent of the angle. (This is the basis of the Box–Muller method for generating normal random numbers.)

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on transformations of random variables.
- **Nassim Nicholas Taleb, *The Black Swan*** (Penguin). Heavy tails and why they matter, for a general reader.
- **Stephen Stigler, *The History of Statistics*** (Harvard University Press). Includes the

Cauchy–Bienaymé dispute.

The full guided solution, and what the question leads to at university, is in Section 12.22:
https://beyondstep.keithmcnulty.org/ch12/cauchy.html